

Supplementary Material for “Message Capacity and Claim Wording Set the Transition Points of Collective Truth-Finding in Language-Model Networks”

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Section numbers, figures, and tables in this document are prefixed with S. References to Sections I–VI, Figures 1–7, and Appendices A–D point to the main article. The glossary (Appendix B), the single-agent coefficient tables and the reduced-map validation (Appendix C), and the fidelity check (Appendix D) are in the main article. In this document and its tables, “threshold” denotes the transition point in the initial fraction (the unstable fixed point), which the main article calls the transition point.

S1 Reanalysis of the public dataset

S1.1 Endogeneity of the initial fraction

Variance decomposition of x_0 and within/between-question outcome correlations (random-graph family, $\delta = 0.1$), computed over the 6,400 objective communities of El et al.:

model	between-question share of $\text{var}(x_0)$	$\text{corr}(x_0, \text{outcome})$	between	within
Gemma-3n	0.995	0.802	0.852	0.041
Llama-3-8B	0.936	0.659	0.721	0.162
Qwen3.5-9B	0.861	0.277	0.351	0.132
GPT-4o-mini	0.503	0.298	0.321	0.273

Question-clustered versus i.i.d. bootstrap intervals for the fitted $q(x_0)$ crossing (random family):

model	crossing	i.i.d. CI	width	question-clustered CI	ratio
GPT-4o-mini	0.494	[0.477, 0.513]	0.036	[0.404, 0.583]	$4.9\times$
Gemma-3n	0.285	[0.251, 0.315]	0.065	[0.163, 0.419]	$3.9\times$
Qwen3.5-9B	0.142	[0.059, 0.206]	0.147	[−1.049, 0.367]	$9.6\times$
Llama-3-8B	0.459	[0.441, 0.480]	0.039	[0.370, 0.564]	$5.0\times$

Two illustrations of the endogeneity: communities at $x_0 \approx 0.40$ on different questions run to opposite consensus with near certainty, while communities at $x_0 = 0.15$ on an easy question converge correctly in 32 of 32 episodes. These are limitations of the identification strategy, not of the dataset’s quality; the same logs support the message-level measurement of the next subsection. The degenerate degree distributions noted in the main article concentrate the random-family mass on $k = 2$ –4, and the lattice family has fixed degree.

S1.2 Message-level coefficients and the truth asymmetry

Question-clustered 95% intervals for the per-message truth asymmetry w_T (43,500 observations per model):

model	w_T	cluster 95% CI	Ising-coupling estimate	contains 1
GPT-4o-mini	1.055	[0.996, 1.114]	1.16	yes (boundary)
Gemma-3n	1.179	[0.731, 1.628]	1.71	yes
Qwen3.5-9B	0.900	[0.759, 1.041]	1.33	yes
Llama-3-8B	1.144	[0.972, 1.316]	1.38	yes

The label attenuation λ_{lab} (per-message weight of “disagree”-labeled edges relative to “agree”-labeled ones) is 0.080 ± 0.044 (GPT-4o-mini), 0.555 ± 0.078 (Gemma), 0.294 ± 0.051 (Qwen), 0.288 ± 0.041 (Llama); θ is nonzero in all four models ($t \geq 3.4$, question-clustered).

S2 Main collective experiment

S2.1 Distribution of final fractions

The outcome labels of the main article ($x_8 \geq 0.6$ correct-side consensus, $x_8 \leq 0.4$ wrong-side) are operational thresholds. The table gives, per campaign, the share of episodes that were decided and the share of decided episodes whose final fraction lies beyond $|x_8 - 0.5| > 0.4$, with the median end state on each side. Pooled over all 2,620 episodes, 66% end in $|x_8 - 0.5| \in [0.45, 0.5]$ (unanimity included).

campaign	episodes	decided ($ x_8 - 0.5 \geq 0.1$)	of which $ x_8 - 0.5 > 0.4$	median x_8 , correct side	median x_8 , wrong side
main arm A ($\rho = 0$)	1008	0.895	0.634	0.922	0.031
main arm A, rewired ($\rho = 1$)	96	0.969	0.667	1.000	0.031
discrimination arm B	300	0.917	0.633	0.875	0.062
true claims (A-1)	288	0.976	0.758	1.000	—
negated rewrites (A-2)	192	1.000	0.995	1.000	—
qwen3:8b, eight claims	256	0.996	0.992	1.000	0.000
qwen3:8b, four claims	480	0.996	0.967	1.000	0.000

S2.2 Outcome family (arm A, generated graphs)

α	$32x_0$	n	q	Wilson 95%	undec.	$q_{\delta=.05}$	$q_{\delta=.2}$	q_{excl}
0.2	4	16	0.000	[0.000, 0.194]	0.000	0.000	0.000	0.000
0.2	8	40	0.025	[0.004, 0.129]	0.000	0.025	0.000	0.025
0.2	12	16	0.000	[0.000, 0.194]	0.000	0.000	0.000	0.000
0.2	16	16	0.000	[0.000, 0.194]	0.000	0.000	0.000	0.000
0.2	20	40	0.100	[0.040, 0.231]	0.000	0.100	0.100	0.100
0.2	24	40	0.275	[0.161, 0.428]	0.000	0.275	0.275	0.275
0.3	4	40	0.000	[-0.000, 0.088]	0.025	0.000	0.000	0.000
0.3	8	16	0.000	[0.000, 0.194]	0.062	0.062	0.000	0.000
0.3	12	16	0.000	[0.000, 0.194]	0.000	0.000	0.000	0.000
0.3	16	16	0.000	[0.000, 0.194]	0.125	0.000	0.000	0.000
0.3	20	40	0.075	[0.026, 0.199]	0.200	0.100	0.050	0.094
0.3	24	40	0.400	[0.263, 0.554]	0.075	0.425	0.400	0.432
0.38	4	16	0.000	[0.000, 0.194]	0.000	0.000	0.000	0.000
0.38	8	16	0.000	[0.000, 0.194]	0.062	0.000	0.000	0.000
0.38	12	16	0.000	[0.000, 0.194]	0.125	0.000	0.000	0.000
0.38	16	40	0.025	[0.004, 0.129]	0.075	0.025	0.000	0.027
0.38	20	40	0.150	[0.071, 0.291]	0.050	0.200	0.150	0.158
0.38	24	40	0.350	[0.221, 0.505]	0.125	0.350	0.325	0.400
0.45	4	40	0.000	[-0.000, 0.088]	0.025	0.000	0.000	0.000
0.45	8	40	0.000	[-0.000, 0.088]	0.150	0.000	0.000	0.000
0.45	12	16	0.000	[0.000, 0.194]	0.125	0.000	0.000	0.000
0.45	16	16	0.000	[0.000, 0.194]	0.125	0.000	0.000	0.000
0.45	20	16	0.000	[0.000, 0.194]	0.312	0.062	0.000	0.000
0.45	24	40	0.450	[0.307, 0.602]	0.075	0.475	0.450	0.486
0.55	4	16	0.000	[0.000, 0.194]	0.062	0.000	0.000	0.000
0.55	8	16	0.000	[0.000, 0.194]	0.125	0.000	0.000	0.000
0.55	12	16	0.000	[0.000, 0.194]	0.125	0.000	0.000	0.000
0.55	16	40	0.100	[0.040, 0.231]	0.175	0.125	0.050	0.121
0.55	20	40	0.225	[0.123, 0.375]	0.175	0.225	0.150	0.273
0.55	24	40	0.375	[0.242, 0.530]	0.075	0.400	0.300	0.405
1	4	16	0.125	[0.035, 0.360]	0.188	0.188	0.125	0.154
1	8	16	0.062	[0.011, 0.283]	0.125	0.125	0.000	0.071
1	12	16	0.062	[0.011, 0.283]	0.312	0.125	0.000	0.091
1	16	40	0.225	[0.123, 0.375]	0.275	0.300	0.150	0.310
1	20	40	0.425	[0.285, 0.578]	0.125	0.450	0.300	0.486
1	24	40	0.400	[0.263, 0.554]	0.275	0.525	0.325	0.552

S2.3 Empirical critical crowding level

α	n (episodes, $x_0 \leq 8/32$)	wrong-consensus rate	claim-cluster 95% CI
0.2	56	0.982	[0.946, 1.000]
0.3	56	0.964	[0.893, 1.000]
0.38	32	0.969	[0.906, 1.000]
0.45	80	0.912	[0.775, 1.000]
0.55	32	0.906	[0.719, 1.000]
1	32	0.750	[0.438, 1.000]

No crossing of the pre-registered 20% threshold occurs in the window (α_{emp}^* undefined; 0 of 2,000 claim-clustered bootstrap resamples cross).

S2.4 Discrimination arm (variant B)

α	n	wrong rate ($x_0 \leq 10/32$)	95% CI	wrong basin retained
0.6	40	0.975	[0.925, 1.000]	True
0.9	40	1.000	[1.000, 1.000]	True
1.3	40	0.975	[0.925, 1.000]	True

Observed low- x_0 ($x_0 \leq 10/32$) mean final fractions against eight-round surrogate rollouts (200 per episode, same graphs and seedings) of three rules with claim-resolved fields: the normalized rule with the 17-claim identified per- k weights, and the per-claim reduced fits with divisive and with constant coupling:

α	observed $\bar{x}_8$	normalized (17-claim weights)	normalized (per-claim fit)	constant (per-claim fit)
0.6	0.062	0.099	0.122	0.246
0.9	0.066	0.133	0.164	0.323
1.3	0.095	0.162	0.198	0.386

The normalized rules lie closer to the observations than the constant rule but neither reproduces them exactly; claim 1569 converged wrong-side under both rules' correct-side predictions (Section IV-D of the main article).

S2.5 Rewiring control

$32x_0$	$n_{\rho=0}$	$q_{\rho=0}$	$n_{\rho=1}$	$q_{\rho=1}$	Δq
4	40	0.000	16	0.000	+0.0000
8	16	0.000	16	0.000	+0.0000
12	16	0.000	16	0.000	+0.0000
16	16	0.000	16	0.000	+0.0000
20	40	0.075	16	0.062	-0.0125
24	40	0.400	16	0.312	-0.0875

S2.6 Yoked one-step calibration

Every realized transition replayed through the response function with the 17-claim identified per- k weights and each claim's own field:

arm	α	transitions	Brier	Brier (climatology)	ECE	residual at $l/k \in [0.4, 0.6]$
A	0.2	43,008	0.095	0.174	0.034	-0.017
A	0.3	67,584	0.108	0.193	0.033	+0.007
A	0.38	43,008	0.120	0.212	0.030	+0.003
A	0.45	43,008	0.112	0.198	0.032	-0.003
A	0.55	43,008	0.127	0.225	0.027	+0.001
A	1	43,008	0.144	0.247	0.030	-0.015
B	0.6	25,600	0.090	0.218	0.027	-0.055
B	0.9	25,600	0.099	0.225	0.027	-0.046
B	1.3	25,600	0.115	0.220	0.035	-0.040

At contested compositions ($l/k \in [0.4, 0.6]$; last column) the residual is within ± 0.02 in arm A, whereas in arm B (own-state line present) a systematic over-prediction of 0.04–0.06 remains.

S3 Polarity discrimination experiment

S3.1 SUPPORTS-claim collectives (A-1)

claim	α	$32x_0 = 4$	$32x_0 = 12$	$32x_0 = 20$	$32x_0 = 28$
993	0.3	0.820	1.000	1.000	1.000
993	0.55	0.974	1.000	1.000	1.000
1103	0.3	0.833	0.875	0.898	0.919
1103	0.55	0.667	0.766	0.784	0.792
1239	0.3	1.000	1.000	1.000	1.000
1239	0.55	1.000	1.000	1.000	1.000

S3.2 Negated-rewrite collectives (A-2) versus originals

rewrite	original	$32x_0 = 4$	$32x_0 = 12$	$32x_0 = 20$	$32x_0 = 28$	orig. mean $\bar{x}_8$ ($x_0 \leq 20/32$)
90006	6	0.984	1.000	1.000	1.000	0.282
90382	382	1.000	1.000	1.000	1.000	0.001
91569	1569	1.000	1.000	1.000	1.000	0.223
92070	2070	1.000	1.000	1.000	1.000	0.029

S3.3 Claim texts, rewrites, and checks

Selected SUPPORTS claims (bare-prompt p_q in parentheses): 993 (0.708) “While there has been a mean rise of a little more than 3mm per year worldwide since the 1990s, in the last decade, the NOAA Virginia Key tide gauge just south of Miami Beach has measured a 9mm rise annually.”; 1103 (0.792) “[ocean acidification was] First referenced in a peer-reviewed study in Nature in 2003”; 1239 (0.792) “Through decades of research, it has become clear that human civilization, roughly 6,000 years old, developed during an unusually stable period for global sea levels.”

Negated rewrites (model opposition-judgment rate; bare-prompt p_q of the rewrite, TRUE = correct side):

original	rewrite	opp. rate	p_q
The polar bear population has been growing.	The polar bear population has <i>not</i> been growing.	0.70	0.344
Sea level rise is decelerating.	Sea level rise is <i>not</i> decelerating.	0.85	0.938
The amount of energy used to construct solar and wind facilities is greater than they produce in their working lives.	...is <i>not</i> greater than they produce in their working lives.	0.70	0.938
When life is considered, ocean acidification is often found to be a non-problem, or even a benefit.	...is often found to be a <i>problem</i> , and <i>not a benefit</i> (contrary; quantifier “often” retained).	0.85	1.000

S3.4 Single-agent predictions for the polarity-experiment cells

Per-claim divisive response functions (Appendix A of the main article) give the predicted phase of each (claim, α) cell of the polarity experiment; all ten cells are predicted monostable correct-side, and all ten converged correct-side. In two cells (993 at $\alpha = 0.30$, 1103 at $\alpha = 0.55$) a minority of episodes seeded at the lowest fraction ended wrong-side, so the fitted 50% crossing lies below the seeding grid; these are the two cells described in the main article as matching to within the resolution of the grid.

claim	α	predicted (per-claim divisive)	observed	obs. $\bar{x}_8$ by $32x_0$
993	0.3	mono-correct	correct from every seeded x_0 (crossing below grid)	0.82 / 1.00 / 1.00 / 1.00
993	0.55	mono-correct	mono-correct	0.97 / 1.00 / 1.00 / 1.00
1103	0.3	mono-correct	mono-correct	0.83 / 0.88 / 0.90 / 0.92
1103	0.55	mono-correct	correct from every seeded x_0 (crossing below grid)	0.67 / 0.77 / 0.78 / 0.79
1239	0.3	mono-correct	mono-correct	1.00 / 1.00 / 1.00 / 1.00
1239	0.55	mono-correct	mono-correct	1.00 / 1.00 / 1.00 / 1.00
90006	0.3	mono-correct	mono-correct	0.98 / 1.00 / 1.00 / 1.00
90382	0.3	mono-correct	mono-correct	1.00 / 1.00 / 1.00 / 1.00
91569	0.3	mono-correct	mono-correct	1.00 / 1.00 / 1.00 / 1.00
92070	0.3	mono-correct	mono-correct	1.00 / 1.00 / 1.00 / 1.00

S4 Cross-model campaigns

S4.1 qwen3:8b: predicted fixed points versus observed collectives

Fixed points of the mean-field map under each claim’s divisive response function (fitted from 336 single-shot queries per claim), with 95% single-agent-bootstrap intervals on the unstable point, against the observed mean final fractions and the fitted 50% crossing:

claim	α	predicted fixed points (s = stable, u = unstable)	pred. x_c [95% CI]	observed $\bar{x}_8$ ($32x_0 = 4/12/20/28$)	obs. crossing [95% CI]
6	0.3	0.0004 (s); 0.2087 (u); 1.0000 (s)	0.21 [0.16, 0.28]	0.200 / 1.000 / 1.000 / 1.000	0.19 [0.10, 0.25]
6	0.55	0.0010 (s); 0.1440 (u); 1.0000 (s)	0.14 [0.09, 0.22]	0.325 / 1.000 / 1.000 / 1.000	0.17 [0.08, 0.23]
6	1	1.0000 (s)	—	0.884 / 1.000 / 1.000 / 1.000	below grid
1239	0.3	0.0000 (s); 0.6994 (u); 0.9998 (s)	0.70 [0.64, 0.75]	0.000 / 0.000 / 0.000 / 1.000	0.75 [0.72, 0.77]
1239	0.55	0.0000 (s); 0.7094 (u); 0.9997 (s)	0.71 [0.64, 0.77]	0.000 / 0.000 / 0.166 / 1.000	0.69 [0.63, 0.75]
1239	1	0.0000 (s); 0.7067 (u); 0.9995 (s)	0.71 [0.64, 0.79]	0.000 / 0.000 / 0.422 / 0.944	0.70 [0.61, 0.79]
1569	0.3	1.0000 (s)	—	1.000 / 1.000 / 1.000 / 1.000	—
1569	0.55	1.0000 (s)	—	1.000 / 1.000 / 1.000 / 1.000	—
1569	1	1.0000 (s)	—	1.000 / 1.000 / 1.000 / 1.000	—
91569	0.3	0.0195 (s); 0.3738 (u); 0.9940 (s)	0.37 [0.22, 0.49]	0.178 / 0.656 / 1.000 / 1.000	0.35 [0.28, 0.43]
91569	0.55	0.9958 (s)	—	0.953 / 1.000 / 1.000 / 1.000	—
91569	1	0.9973 (s)	—	1.000 / 1.000 / 1.000 / 1.000	—

Per-cell outcome counts (correct/wrong/undecided of 10 episodes) for the same campaign; intermediate mean final fractions in Fig. 6 of the main article are splits between the two basins:

claim	α	$32x_0 = 4$	$32x_0 = 12$	$32x_0 = 20$	$32x_0 = 28$
6	0.3	2/8/0	10/0/0	10/0/0	10/0/0
6	0.55	3/7/0	10/0/0	10/0/0	10/0/0
6	1	9/1/0	10/0/0	10/0/0	10/0/0
1239	0.3	0/10/0	0/10/0	0/10/0	10/0/0
1239	0.55	0/10/0	0/10/0	2/8/0	10/0/0
1239	1	0/10/0	0/10/0	3/7/0	9/0/1
1569	0.3	10/0/0	10/0/0	10/0/0	10/0/0
1569	0.55	10/0/0	10/0/0	10/0/0	10/0/0
1569	1	10/0/0	10/0/0	10/0/0	10/0/0
91569	0.3	0/9/1	6/4/0	10/0/0	10/0/0
91569	0.55	10/0/0	10/0/0	10/0/0	10/0/0
91569	1	10/0/0	10/0/0	10/0/0	10/0/0

qwen3:8b calibration of the 120 candidate claims found 4 SUPPORTS and 3 REFUTES claims with $|h_q| < 0.4$; a polarity-balanced set of 12 could not be formed, so genuine truth asymmetry remains unidentified in this model as well.

S4.2 llama3.3:70b single-agent probe

claim	c_0	β_T	β_F	n	$P(\text{TRUE})$ at $k = 0$
6	1.987	0.636	-0.257	132	0.000
382	3.920	16.536	-0.500	132	0.000
1569	36.547	5.377	-4.482	132	0.000
2070	3.455	1.375	-0.554	132	0.000
90006	0.378	0.487	-0.494	132	0.417
90382	1.739	1.407	-0.748	132	1.000
91569	1.713	1.156	-0.572	132	1.000
92070	3.910	0.203	0.203	132	1.000

Polarity-coded regression across all eight claims (claim fixed effects): $b(\text{TRUE-side message}) = +0.583$, $b(\text{FALSE-side}) = -0.563$, ratio 1.03 with claim-clustered 95% interval $[0.66, 1.64]$ (2,000 resamples of the eight claims; weak ridge 10^{-3} against separation). On the same eight claims and message bank llama3.1:8b gives 1.94 $[1.39, 2.73]$ (over the 17 calibration claims: 1.44 $[1.25, 1.65]$); the 8b–70B difference is 0.90 $[0.00, 1.72]$ and the ratio of ratios 1.87 $[1.00, 3.12]$, so the reduction is at the edge of resolution with eight claims. Per-claim coefficients with $|c_0| > 5$ in the table above reflect separation (accuracy 0 or 1 at $k = 0$) and are reported for completeness. Empty-inbox fields (last column, observed rates over the empty-inbox queries of each claim): the four false originals give $P(\text{TRUE}) = 0.000$ (every answer on the denying and correct side); of the four true rewrites, three give 1.000 and the negated polar-bear rewrite (90006) 0.42, the one near-neutral claim among the eight. Per- k fits (truth-coded, pooled across claims): $\beta_T(4) = 2.41 \rightarrow \beta_T(8) = 1.35$. The positive wrong-side coefficient at $k = 4$ (+0.61) — wrong-side messages slightly *increasing* correct-side probability — is recorded as an observation only.

S5 Stage 2c: eight-claim calibration campaign and polarity-origin probe

S5.1 Calibration campaign: all 16 cells

Predicted classifications and thresholds come from each claim’s divisive response function fitted to its single-shot queries (Appendix A of the main article); the eight claims were selected from twenty candidates by the pre-registered rule. Prediction intervals are single-agent bootstrap percentiles (200 resamples of each claim’s queries within design cells) and “frac. bistable” is the fraction of resamples classified bistable. Observed crossings are 50% points of ridge-penalized logistic fits of episode outcomes against x_0 , with episode-bootstrap 95% intervals; $n_c/n_w/n_u$ are correct/wrong/undecided episode counts of 16 per cell.

S5.2 Polarity-origin probe (D-3)

Judgment-side single-shot measurements on three llama3.1-8b variants with the fixed 8B message bank and a three-shot answer-format prefix applied identically to all variants. The base model does not follow the forced-choice format (parse rate 1.5% in the bare pilot, 4.9% under the prefix), so its row is reported for completeness and carries no statistical weight.

claim	α	predicted	pred. x_c [95% CI] (frac. bistable)	observed	obs. crossing [95% CI]	$n_c/n_w/n_u$
97	0.3	mono-correct	—	mono-correct	—	16/0/0
97	0.55	mono-correct	—	mono-correct	—	16/0/0
184	0.3	mono-correct	—	mono-correct	—	16/0/0
184	0.55	mono-correct	—	mono-correct	—	16/0/0
590	0.3	bistable	0.559 [0.497, 0.619] (1.00)	bistable	0.560 [0.465, 0.653]	7/9/0
590	0.55	bistable	0.578 [0.516, 0.637] (1.00)	bistable	0.622 [0.536, 0.701]	6/10/0
656	0.3	bistable	0.185 [0.125, 0.267] (1.00)	bistable	0.248 [0.157, 0.341]	12/4/0
656	0.55	bistable	0.119 [0.072, 0.187] (1.00)	bistable	0.120 [0.057, 0.200]	14/2/0
832	0.3	bistable	0.425 [0.364, 0.494] (1.00)	bistable	0.498 [0.413, 0.589]	8/8/0
832	0.55	bistable	0.396 [0.331, 0.466] (1.00)	bistable	0.436 [0.334, 0.531]	9/7/0
1013	0.3	bistable	0.407 [0.344, 0.470] (1.00)	bistable	0.436 [0.335, 0.525]	9/7/0
1013	0.55	bistable	0.409 [0.346, 0.471] (1.00)	bistable	0.373 [0.292, 0.460]	10/6/0
1146	0.3	bistable	0.094 [0.046, 0.157] (0.98)	mono-correct	—	16/0/0
1146	0.55	bistable	0.034 [0.025, 0.101] (0.62)	bistable	0.012 [-0.087, 0.169]	15/1/0
1151	0.3	bistable	0.271 [0.147, 0.395] (0.96)	bistable	0.184 [0.085, 0.285]	13/3/0
1151	0.55	bistable	0.288 [0.178, 0.407] (0.95)	bistable	0.311 [0.208, 0.404]	11/4/1

Table S1: All 16 cells of the eight-claim calibration campaign (qwen3:8b, 256 episodes, zero parse failures). Fifteen cells match in class; the mismatch is claim 1146 at $\alpha = 0.30$, predicted bistable with threshold 0.094 just above the lowest seeded fraction (2/32), observed monostable correct. All eleven predicted thresholds of the cells observed bistable lie inside the observed intervals (mean absolute difference 0.04; Spearman $r_s = 0.96$, claim-permutation $p = 0.004$).

variant	tag (digest)	parsed	$ b_T / b_F $	95% CI
instruct (control)	llama3.1:8b (46e0c10c039e)	5,984/5,984	1.437	[1.250, 1.651]
abliterated	mannix/llama3.1-8b-abliterated (f7d729db9832, Q4_0)	5,984/5,984	1.243	[1.051, 1.471]
base	llama3.1:8b-text-q4_K_M (6f98b5a6e4b7)	294/5,984	(2.10)	not evaluable

Table S2: Per-message assertion asymmetry (claim-fixed-effect logistic regression; claim-clustered bootstrap intervals). The instruct control reproduces the Stage-1 value 1.44 under the added format prefix. Bare affirmation indices S for the four original/negated pairs: instruct 0.92–1.52 (mean 1.22), abliterated 1.13–1.42 (mean 1.24); base cells parsed 2–6 of 64 and are not evaluable.

S6 Residual and consistency analyses

S6.1 Message statistics by side

claim	side	messages	words/msg	hedges/msg
6	correct	28258	60.7	0.058
6	wrong	35998	59.4	0.419
382	correct	16713	63.5	0.035
382	wrong	47591	62.9	0.116
1569	correct	28115	58.8	0.084
1569	wrong	36093	59.6	0.314
2070	correct	14105	64.5	0.081
2070	wrong	50119	59.8	1.494

S6.2 Recorded-message persuasiveness probes

claim	side	message source	$P(\text{correct})$	n
6	correct	bank	1.000	60
6	correct	episode	1.000	60
6	wrong	bank	0.100	60
6	wrong	episode	0.167	60
382	correct	bank	1.000	60
382	correct	episode	1.000	60
382	wrong	bank	0.017	60
382	wrong	episode	0.000	60
1569	correct	bank	1.000	60
1569	correct	episode	1.000	60
1569	wrong	bank	0.100	60
1569	wrong	episode	0.050	60
2070	correct	bank	1.000	60
2070	correct	episode	1.000	60
2070	wrong	bank	0.050	60
2070	wrong	episode	0.167	60

Episode-generated and bank messages are indistinguishable in persuasive force; no message-quality contribution to the high- x_0 residual is detected.

S6.3 Repeated exposure and finite time

One-step regression with round interactions gives $\beta_T \times (t - 1) = +0.026$ (SE 0.006) and $\beta_F \times (t - 1) = -0.004$ (SE 0.003): repeated exposure does not attenuate message influence. Finite-time check on high- x_0 episodes:

α	$32x_0$	n	mean $x_8 - x_7$	frac. interior	frac. still moving
0.2	20	40	-0.008	0.300	0.275
0.2	24	40	-0.009	0.225	0.150
0.3	20	40	-0.012	0.450	0.250
0.3	24	40	+0.001	0.275	0.325
0.38	20	40	+0.001	0.450	0.375
0.38	24	40	-0.028	0.375	0.350
0.45	20	16	-0.020	0.438	0.375
0.45	24	40	-0.008	0.275	0.375
0.55	20	40	-0.009	0.525	0.375
0.55	24	40	-0.002	0.450	0.225
1	20	40	-0.002	0.800	0.575
1	24	40	-0.027	0.725	0.550

S6.4 Label residual

Correlation between an episode’s share of TF label assignments and its final state: $r = 0.194$ ($n = 1,008$ episodes) — an order of magnitude below the experimental effects ($\Delta \bar{x}_8 \approx 1$), but nonzero; per-call randomization protects pooled estimates.

S6.5 Flip statistics and cascades

Refractory regression: the coefficient of “flipped at $t-1$ ” on flipping at t is $+0.98$ (claim-cluster SE 0.13) — flips concentrate in particular agents (anti-refractory). Branching ratios:

α	flip events	σ (mean downstream flips)
0.2	9048	3.546
0.3	10129	3.102
0.38	11452	2.771
0.45	10473	2.476
0.55	12301	2.312
1	14625	1.776

Cascade-size distribution in the asynchronous public dataset (GPT-4o-mini, 320 runs; cascades linked by graph adjacency within three events):

cascade size	count
1	9341
2	668
3	130
4	28
5	9
6	1
7	1

91.8% of cascades have size 1; the tail decays geometrically (deeply subcritical).

S7 Extended Methods

This section reproduces in full the Methods subsections that Appendix A of the main article summarizes: models and serving, verbatim prompts, the cross-model calibration campaign, the complete statistical procedures, and the pre-registration statement.

S7.1 Models and serving

All agents are served with ollama. The primary model is llama3.1:8b (Q4_K_M quantization, temperature 0.7 throughout, seed recorded for every call). The cross-model campaign uses qwen3:8b (tag 500a1f067a9f, thinking disabled via `think:false`) and llama3.3:70b (Q4_K_M; judgments only, message banks fixed to the 8B-generated banks). Single-shot campaigns ran against a user-space ollama instance with `OLLAMA_NUM_PARALLEL=16` (the system instance saturates at 0.62 generations/s; the parallel instance reaches 3.5 generations/s); the 70B model does not load under 16-slot KV allocation and was served by the serial system instance. The polarity-origin probe (Section V of the main article) used two further variants, judgments only: llama3.1:8b-text-q4_K_M (the base model) and an ablated derivative of the instruct model (mannix/llama3.1-8b-ablated, Q4_0), with a three-shot answer-format prefix applied identically to all variants, including the re-measured instruct control. Every call’s prompt hash, seed, raw response, and parse result are logged to JSONL.

S7.2 Prompts

The judgment prompt (verbatim; variant A) is:

```
# Your Expertise
{persona}
# Messages
You received the following messages from fellow panelists:
- {message 1} ...
```

```

# Problem
You are one member of an expert panel judging the following claim.
Claim: "{claim}"
Options:
A) {TRUE or FALSE}
B) {FALSE or TRUE}
# Instruction
Give your direct answer on which option is correct.
Do NOT work through the problem, show any steps, compute, or
explain --- just commit to a snap judgement.
## Response Format
Respond with a single character, A or B, and nothing else.

```

Variant B inserts, before # Problem: “# Your Current Answer / Your current answer is that the claim is {TRUE|FALSE}.” The A/B $\leftrightarrow$ TRUE/FALSE assignment is randomized per call and counterbalanced within cells; message presentation order is randomized and recorded. The message-generation prompt asks the sender, given its persona, the claim, and its current stance, to “write a brief message to a fellow panelist stating that you believe the claim is {stance} and giving the single strongest reason from your reasoning,” with a two-sentence response format; stance is expressed in the message body (TRUE/FALSE), never as an option letter, because option labels are randomized per receiver. Personas are one-line expertise statements drawn from a fixed pool of 12. Prompt texts are extracted verbatim from the released scripts.

S7.3 Cross-model calibration campaign

Twenty further CLIMATE-FEVER claims (ten SUPPORTS, ten REFUTES, disjoint from all earlier sets) were calibrated under the full scaffold and measured single-shot with qwen3:8b (24 calls per candidate, 64 in the boundary band; per-claim response-function fits from 6,720 further queries). The resulting fixed-point classifications and thresholds for $\alpha \in \{0.30, 0.55\}$ were computed at the time of the campaign, and eight claims were selected by a rule fixed in the pre-registered instructions (at least four predicted bistable with interior thresholds, the remainder monostable, filled from the bistable pool where a category was empty — no wrong-side-dominant claim existed among the twenty candidates). Collectives ran the full grid $\alpha \in \{0.30, 0.55\} \times x_0 \in \{2, 6, \dots, 30\}/32$ with two replicates (256 episodes, protocol identical to the main cross-model campaign). Observed thresholds are defined under Statistics (Appendix A of the main article) and in the next subsection.

S7.4 Statistics: full procedures

All confidence intervals on measured coefficients are claim-clustered pairs bootstrap percentiles (1,000–2,000 resamples for the pooled coefficients, 200 for the identified per- k coefficients and for the prediction-side intervals, 2,000 for the polarity ratios; claims are the clusters because episodes within a claim share its field). Attenuation-form comparison uses five-fold cross-validation with folds split by claim, alongside AIC; the form parameter (γ for divisive normalization, the rate for load decay) is profiled on the full data ($\gamma = 0.9$) and re-profiled inside each training fold, and a held-out claim’s fixed effect, which cannot be estimated from the training folds, is set to zero, so the cross-validated likelihood scores prediction for an unseen claim with an unknown field. The divisive α^* of Section III of the main article uses γ profiled separately per variant (0.53 for A, 1.32 for B); Fig. 3 of the main article uses the pooled $\gamma = 0.9$. Per- k coefficients are identified by a joint logistic fit over all inbox sizes with claim fixed effects anchored by the empty-inbox cells: at a single k the columns 1, l , and $k - l$ are collinear, so separate per- k fits identify only $c_0 + k\beta_F$ and $\beta_T - \beta_F$. Per-claim response functions for the collective claims use the reduced form $c_0 + g(k)[\delta k + \bar{\beta}(2l - k)]$

($+\theta s$ for variant B), with a weak ridge penalty (10^{-3}) where k takes three values. For Fig. 3 of the main article, one logistic fit per prompt variant with claim fixed effects and message weights $\beta_{Tg}(k)$, $\beta_{Fg}(k)$ (divisive, $\gamma = 0.9$ fixed, or $g \equiv 1$) assigns each query a predicted probability; a design cell’s net input u is the logit of its mean predicted probability, and its observed correct-side rate is plotted against u with a Wilson 95% interval. Fixed points of the mean-field map are bracketed on an endpoint-inclusive grid of 4,001 points in x with bisection refinement and classified by $|F'(x)| < 1$; a phase is bistable when an unstable interior point separates two stable ones, and α^* is located by bisection in α on that classification, with the fold of G solved jointly from $G = x$ and $G_x = 1$. Prediction-side intervals resample each claim’s single-shot outcomes within that claim and within each design cell (200 resamples; a within-claim bootstrap, not the claim-clustered bootstrap used for pooled coefficients) and recompute the fixed points. Reanalysis statistics on the public dataset use question-clustered bootstrap and CR0 sandwich errors.

Outcome and threshold. An episode’s outcome is its correct-side fraction after eight rounds, x_8 ; $q(x_0)$ is the fraction of episodes with $x_8 \geq 0.6$, a finite-horizon quantity that we compare with the infinite-time fixed points only through surrogate rollouts. Observed thresholds are the 50% crossings of a logistic fit of the correct outcome on x_0 with a weak ridge penalty (0.002) so that separated cells have a finite estimate; the slope is not constrained, a fit with a non-positive slope would be discarded, and none occurred in the reported cells. Intervals are episode-bootstrap percentiles (500 replicates); replicates without a finite crossing in $(-1, 2)$ are dropped, which retained all replicates in 12 of the 17 cells observed bistable, 97–99% in three, and 89% and 65% in the two cells whose crossing lies nearest the lowest seeded fraction (claims 656 and 1146 at $\alpha = 0.55$). Rank correlations between predicted and observed thresholds are tested by permuting claim labels, claims being the exchangeable unit.

S7.5 Pre-registration timeline: full statement

Throughout this paper, “pre-registered” means that a prediction or a decision rule, together with the criteria for judging it, was written to a file before the corresponding data were collected, and that the file’s timestamp and SHA-256 hash are recorded in the project repository, so that the order of prediction and data is documented; these are local records, not a third-party time stamp. No external registry (such as OSF) was used. Three items were pre-registered: (i) the $\alpha^* = 0.435 \pm 0.03$ prediction and the decision criteria of the main collective experiment, before any collective episode (2026-08-23); (ii) the three-hypothesis prediction table of the polarity discrimination experiment, before its data collection (2026-08-25); (iii) the claim-selection rule of the eight-claim calibration campaign, written into the campaign instructions before its data were collected (2026-08-27). The single-agent measurements preceded the collective experiments. The claim-resolved predictions of Sections IV-B to IV-D of the main article were computed after the collective experiments, with the estimation procedure described under Statistics, using only the single-agent data for parameter fitting; they were not prospectively registered. The one adaptive element (second-pass allocation of episodes) followed rules fixed in the pre-registered design.

References